# Do Now · retrieval

## Thinking abstractly: models and reality

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is unit testing?

### Q2

What is a precondition?

### Q3

What do vertices and edges represent in a graph?

### Q4

Why is an unweighted route graph not enough to identify the fastest journey?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Choose a purpose — Decide whether a particular book may be loaned.

## Core idea 2

Keep essential facts — ISBN identifies the book, and Available indicates whether it can be borrowed.

## Core idea 3

Evaluate omissions — The model cannot answer questions about book weight or appearance because those attributes are omitted.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

abstraction — Retain relevant characteristics, ignore irrelevant ones.

model — Simplified representation serving a purpose.

attribute — Property stored about an entity.

assumption — Condition accepted in constructing the model.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

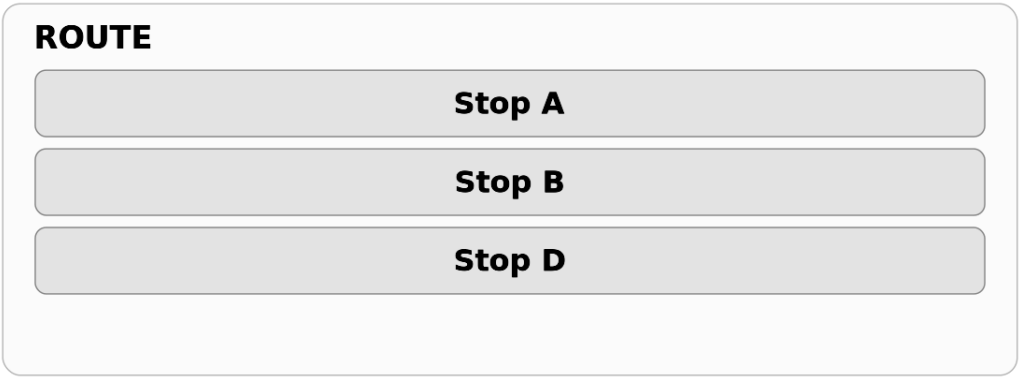
Model a bus network for finding the route with the fewest stops.

## Follow the worked method

1. Predict first — Model each bus stop as a vertex and each permitted connection as an edge.

2. Model and check — Vehicle colour is irrelevant. To find fastest rather than fewest-stop routes, add journey-time weights.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first

Only valid marks and the number of entries are needed to compute the mean.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Identify two relevant and two irrelevant attributes for a stock checker.

## Question 2

Give a simple abstraction for detecting school room timetable clashes: three needed fields and two omitted details.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Why should a model not contain every real-world detail? Give an example where a previously omitted detail becomes relevant.

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

A good abstraction must record every real-world property.

## Use key vocabulary from memory

1. Explain abstraction in your own words.

2. Explain model in your own words.

3. Explain attribute in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.